

Seat No:

SET Q

AUTONOMOUS STATE EXAMINATION BOARD**Btech Computer Scie Engg DBMS****Subject: Data Structure · Pattern: CBCS Pattern****Day & Date:** Saturday, September 19, 2026 **Duration:** 180 Minutes **Max. Marks:** 70 Marks**Instructions for Candidates:**

1. Q.1 is compulsory. Mention the question paper set (**Q**) clearly on top of the answer booklet.
2. Figures to the right indicate full marks assigned to each question.
3. Assume suitable data wherever necessary and state your assumptions explicitly.
4. Use of programmable calculators or unauthorized electronic communication devices is strictly prohibited.
5. Standard University Marking Scheme

Q.1 Choose the correct alternatives for the following questions. [14 Marks]

1. What is the minimum number of scalar multiplications required to multiply a chain of matrices with dimensions: A1 (10x30), A2 (30x5), A3 (5x60)?
 - a) 4,500
 - b) 2,700
 - c) 18,000
 - d) 3,000
2. In open addressing with linear probing, what phenomenon occurs when filled hash slots form continuous clusters, degrading search performance?
 - a) Secondary Clustering
 - b) Primary Clustering
 - c) Hash Overflow
 - d) Perfect Hashing
3. What is the worst-case time complexity of searching an element in an AVL Tree with n nodes?
 - a) $O(1)$
 - b) $O(\log n)$
 - c) $O(n)$
 - d) $O(n \log n)$
4. Which of the following sorting algorithms is inherently stable and operates with $O(n \log n)$ worst-case time complexity?
 - a) Quick Sort
 - b) Heap Sort
 - c) Merge Sort

- d) Selection Sort
5. Which CPU scheduling algorithm gives the minimum average waiting time for a given set of processes?
- a) First-Come First-Served (FCFS)
 - b) Shortest Job First (SJF / Preemptive SRTF)
 - c) Round Robin (RR)
 - d) Priority Scheduling
6. Which normal form is based on the concept of full functional dependency and eliminates partial dependency on candidate keys?
- a) First Normal Form (1NF)
 - b) Second Normal Form (2NF)
 - c) Third Normal Form (3NF)
 - d) Boyce-Codd Normal Form (BCNF)
7. Belady's Anomaly where allocating more page frames results in more page faults can occur in which page replacement algorithm?
- a) Optimal Page Replacement (OPT)
 - b) Least Recently Used (LRU)
 - c) First-In First-Out (FIFO)
 - d) Least Frequently Used (LFU)
8. Which algorithm is utilized by operating systems to safely avoid deadlocks when processes declare their maximum resource claims in advance?
- a) Banker's Algorithm
 - b) Peterson's Algorithm
 - c) Lamport's Bakery Algorithm
 - d) Dekker's Algorithm
9. Which relational algebra operation selects rows from a relation that satisfy a specified predicate or condition?
- a) Projection (σ)
 - b) Selection (π)
 - c) Cartesian Product ($\times$)
 - d) Natural Join ($\bowtie$)
10. Which algorithm finds the single-source shortest paths in a directed graph containing edges with negative weights (assuming no negative weight cycles)?
- a) Dijkstra's Algorithm
 - b) Bellman-Ford Algorithm
 - c) Prim's Algorithm
 - d) Kruskal's Algorithm

11. In database transaction management, which ACID property guarantees that concurrent execution of transactions results in a state equivalent to serial execution?
- a) Atomicity
 - b) Consistency
 - c) Isolation
 - d) Durability
12. In a B+ Tree of order p , where are the actual data record pointers (or records) stored?
- a) Only in the internal nodes
 - b) Only in the leaf nodes
 - c) Uniformly in both leaf and internal nodes
 - d) At the root node only
13. Which locking protocol prevents cascading aborts in concurrent transaction execution?
- a) Basic Two-Phase Locking (2PL)
 - b) Strict Two-Phase Locking (Strict 2PL)
 - c) Conservative Two-Phase Locking
 - d) Graph-based Tree Locking Protocol
14. Which normal form is based on the concept of full functional dependency and eliminates partial dependency on candidate keys?
- a) First Normal Form (1NF)
 - b) Second Normal Form (2NF)
 - c) Third Normal Form (3NF)
 - d) Boyce-Codd Normal Form (BCNF)

SECTION – I (Theory & Core Concepts)

[28 Marks]

Q.2 Answer the following questions (Attempt Any Four):

[16 Marks]

- a) Explain Dijkstra's shortest path algorithm using a min-priority queue (Fibonacci Heap). Provide a formal proof of correctness by induction and analyze its time complexity as $O(E + V \log V)$. [10]
- b) Formulate the dynamic programming recurrence relation for the 0/1 Knapsack Problem with capacity W and items of weights w_i and values v_i . Analyze its time and space complexity. [5]
- c) Describe the ARIES recovery algorithm in database management systems. Detail the three phases: Analysis, Redo, and Undo, along with the significance of Write-Ahead Logging (WAL) and checkpoints. [10]
- d) Explain the two-level hierarchical paging scheme with Translation Lookaside Buffer (TLB). Calculate the Effective Memory Access Time (EMAT) if TLB hit ratio is 95%, TLB lookup time is 20 ns, and main memory access time is 100 ns. [10]

Q.3 Answer the following questions in detail (Attempt Any Two): [12 Marks]

- a) Explain Dijkstra's shortest path algorithm using a min-priority queue (Fibonacci Heap). Provide a formal proof of correctness by induction and analyze its time complexity as $O(E + V \log V)$. [10]
 - b) Formulate the dynamic programming recurrence relation for the 0/1 Knapsack Problem with capacity W and items of weights w_i and values v_i . Analyze its time and space complexity. [5]
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SECTION – II (Analysis, Design & Applications) [28 Marks]**Q.4 Answer the following questions (Attempt Any Four):** [16 Marks]

- a) State the four Coffman conditions necessary for a deadlock to occur in an operating system. Explain how preventing the Circular Wait condition avoids system deadlock. [5]
- b) Explain the min-heap property. Illustrate step-by-step the procedure to build a binary min-heap from the unsorted array [12, 11, 13, 5, 6, 7] using bottom-up heapification. [5]
- c) Explain the difference between clustered and non-clustered indexes in SQL databases. Discuss why a table can have only one clustered index while having multiple non-clustered indexes. [5]
- d) Given a relational schema $R(A, B, C, D, E)$ with functional dependencies $F = A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A$. Find all candidate keys of R and determine the highest normal form satisfied by R . [5]

Q.5 Solve / Explain the following technical problems: [12 Marks]

- a) State the four Coffman conditions necessary for a deadlock to occur in an operating system. Explain how preventing the Circular Wait condition avoids system deadlock. [5]
- b) Explain the min-heap property. Illustrate step-by-step the procedure to build a binary min-heap from the unsorted array [12, 11, 13, 5, 6, 7] using bottom-up heapification. [5]

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